

Divided Differences

Divided Differences

Suppose that $P_n(x)$ is the n th Lagrange polynomial that agrees with the function f at the distinct numbers $x_0, x_1, \dots, x_n$. Although this polynomial is unique, there are alternate algebraic representations that are useful in certain situations. The divided differences of f with respect to $x_0, x_1, \dots, x_n$ are used to express $P_n(x)$ in the form

$$P_n(x) = a_0 + a_1(x - x_0) + a_2(x - x_0)(x - x_1) + \dots + a_n(x - x_0) \dots (x - x_{n-1}), \quad (3.5)$$

for appropriate constants $a_0, a_1, \dots, a_n$. To determine the first of these constants, a_0 , note that if $P_n(x)$ is written in the form of Eq. (3.5), then evaluating $P_n(x)$ at x_0 leaves only the constant term a_0 ; that is,

$$a_0 = P_n(x_0) = f(x_0).$$

Similarly, when $P(x)$ is evaluated at x_1 , the only nonzero terms in the evaluation of $P_n(x_1)$ are the constant and linear terms,

$$f(x_0) + a_1(x_1 - x_0) = P_n(x_1) = f(x_1);$$

so

$$a_1 = \frac{f(x_1) - f(x_0)}{x_1 - x_0}.$$

We now introduce the divided-difference notation

The *zeroth divided difference* of the function f with respect to x_i , denoted $f[x_i]$, is simply the value of f at x_i :

$$f[x_i] = f(x_i).$$

The remaining divided differences are defined recursively; the *first divided difference* of f with respect to x_i and x_{i+1} is denoted $f[x_i, x_{i+1}]$ and defined as

$$f[x_i, x_{i+1}] = \frac{f[x_{i+1}] - f[x_i]}{x_{i+1} - x_i}.$$

The *second divided difference*, $f[x_i, x_{i+1}, x_{i+2}]$, is defined as

$$f[x_i, x_{i+1}, x_{i+2}] = \frac{f[x_{i+1}, x_{i+2}] - f[x_i, x_{i+1}]}{x_{i+2} - x_i}.$$

Similarly, after the $(k - 1)$ st divided differences,

$$f[x_i, x_{i+1}, x_{i+2}, \dots, x_{i+k-1}] \quad \text{and} \quad f[x_{i+1}, x_{i+2}, \dots, x_{i+k-1}, x_{i+k}],$$

have been determined, the **k th divided difference** relative to $x_i, x_{i+1}, x_{i+2}, \dots, x_{i+k}$ is

$$f[x_i, x_{i+1}, \dots, x_{i+k-1}, x_{i+k}] = \frac{f[x_{i+1}, x_{i+2}, \dots, x_{i+k}] - f[x_i, x_{i+1}, \dots, x_{i+k-1}]}{x_{i+k} - x_i}.$$

The process ends with the single *n th divided difference*,

$$f[x_0, x_1, \dots, x_n] = \frac{f[x_1, x_2, \dots, x_n] - f[x_0, x_1, \dots, x_{n-1}]}{x_n - x_0}.$$

$$\begin{aligned} P_n(x) = & f[x_0] + f[x_0, x_1](x - x_0) + a_2(x - x_0)(x - x_1) \\ & + \dots + a_n(x - x_0)(x - x_1) \dots (x - x_{n-1}). \end{aligned}$$

As might be expected from the evaluation of a_0 and a_1 , the required constants are

$$a_k = f[x_0, x_1, x_2, \dots, x_k],$$

for each $k = 0, 1, \dots, n$. So $P_n(x)$ can be rewritten in a form called Newton's Divided-Difference:

$$P_n(x) = f[x_0] + \sum_{k=1}^n f[x_0, x_1, \dots, x_k](x - x_0) \dots (x - x_{k-1}).$$

x	$f(x)$	First divided differences	Second divided differences	Third divided differences
x_0	$f[x_0]$			
		$f[x_0, x_1] = \frac{f[x_1] - f[x_0]}{x_1 - x_0}$		
x_1	$f[x_1]$		$f[x_0, x_1, x_2] = \frac{f[x_1, x_2] - f[x_0, x_1]}{x_2 - x_0}$	
		$f[x_1, x_2] = \frac{f[x_2] - f[x_1]}{x_2 - x_1}$		$f[x_0, x_1, x_2, x_3] = \frac{f[x_1, x_2, x_3] - f[x_0, x_1, x_2]}{x_3 - x_0}$
x_2	$f[x_2]$		$f[x_1, x_2, x_3] = \frac{f[x_2, x_3] - f[x_1, x_2]}{x_3 - x_1}$	
		$f[x_2, x_3] = \frac{f[x_3] - f[x_2]}{x_3 - x_2}$		$f[x_1, x_2, x_3, x_4] = \frac{f[x_2, x_3, x_4] - f[x_1, x_2, x_3]}{x_4 - x_1}$
x_3	$f[x_3]$		$f[x_2, x_3, x_4] = \frac{f[x_3, x_4] - f[x_2, x_3]}{x_4 - x_2}$	
		$f[x_3, x_4] = \frac{f[x_4] - f[x_3]}{x_4 - x_3}$		$f[x_2, x_3, x_4, x_5] = \frac{f[x_3, x_4, x_5] - f[x_2, x_3, x_4]}{x_5 - x_2}$
x_4	$f[x_4]$		$f[x_3, x_4, x_5] = \frac{f[x_4, x_5] - f[x_3, x_4]}{x_5 - x_3}$	
		$f[x_4, x_5] = \frac{f[x_5] - f[x_4]}{x_5 - x_4}$		
x_5	$f[x_5]$			

$$f[\underbrace{x_i, x_{i+1}, \dots, x_{i+k}}] = \frac{f[x_{i+1}, \dots, x_{i+k}] - f[x_i, \dots, x_{i+k-1}]}{x_{i+k} - x_i}$$

A polynomial of order 2 that passes through the three points

$(x_0, f(x_0)), (x_1, f(x_1)), (x_2, f(x_2)).$

$$p_2(x) = f[x_0] + f[x_0, x_1](x - x_0) + f[x_0, x_1, x_2](x - x_0)(x - x_1)$$

x	$f(x)$	First divided differences	Second divided differences
x_0	$f[x_0]$		
		$f[x_0, x_1] = \frac{f[x_1] - f[x_0]}{x_1 - x_0}$	
x_1	$f[x_1]$		$f[x_0, x_1, x_2] = \frac{f[x_1, x_2] - f[x_0, x_1]}{x_2 - x_0}$
		$f[x_1, x_2] = \frac{f[x_2] - f[x_1]}{x_2 - x_1}$	
x_2	$f[x_2]$		

The coefficients of the Newton divided-difference form of the interpolating polynomial are along the **diagonal** in the table.

General Form of Newton's Interpolating Polynomials

A polynomial of order n that passes through the $(n+1)$ points

$$(x_0, f(x_0)), (x_1, f(x_1)), \dots, (x_n, f(x_n)).$$

$$p_n(x) = f[x_0] + f[x_0, x_1](x - x_0) + f[x_0, x_1, x_2](x - x_0)(x - x_1) \\ + \dots + f[x_0, x_1, \dots, x_n](x - x_0)(x - x_1) \dots (x - x_{n-1})$$

x_n does not appear in the product

Example

Fit a second-order polynomial to the three points

x_i	$f[x_i]$
$x_0=1$	$f[x_0]=1$
$x_1=2$	$f[x_1]=8$
$x_2=4$	$f[x_2]=14$

x_i	$f[x]$	first divided difference	second divided difference
$x_0=1$	$f[x_0]=1$	$f[x_0, x_1] = \frac{8-1}{2-1} = 7$	$f[x_0, x_1, x_2] = \frac{f[x_1, x_2] - f[x_0, x_1]}{x_2 - x_0} = \frac{3-7}{4-1} = -\frac{4}{3}$
$x_1=2$	$f[x_1]=8$		
$x_2=4$	$f[x_2]=14$	$f[x_1, x_2] = \frac{14-8}{4-2} = 3$	

$$p_2(x) = f[x_0] + f[x_0, x_1](x - x_0) + f[x_0, x_1, x_2](x - x_0)(x - x_1)$$

$$p_2(x) = 1 + 7(x - 1) - \frac{4}{3}(x - 1)(x - 2)$$

Example Complete the divided difference table for the data and construct the interpolating polynomial that uses all this data.

x	$f(x)$
1.0	0.7651977
1.3	0.6200860
1.6	0.4554022
1.9	0.2818186
2.2	0.1103623

Solution The first divided difference involving x_0 and x_1 is

$$f[x_0, x_1] = \frac{f[x_1] - f[x_0]}{x_1 - x_0} = \frac{0.6200860 - 0.7651977}{1.3 - 1.0} = -0.4837057.$$

The remaining first divided differences are found in a similar manner and are shown in the fourth column in Table 3.11.

i	x_i	$f[x_i]$	$f[x_{i-1}, x_i]$	$f[x_{i-2}, x_{i-1}, x_i]$	$f[x_{i-3}, \dots, x_i]$	$f[x_{i-4}, \dots, x_i]$
0	1.0	0.7651977				
1	1.3	0.6200860	-0.4837057			
2	1.6	0.4554022	-0.5489460	-0.1087339	0.0658784	
3	1.9	0.2818186	-0.5786120	-0.0494433	0.0680685	0.0018251
4	2.2	0.1103623	-0.5715210	0.0118183		

The second divided difference involving x_0 , x_1 , and x_2 is

$$f[x_0, x_1, x_2] = \frac{f[x_1, x_2] - f[x_0, x_1]}{x_2 - x_0} = \frac{-0.5489460 - (-0.4837057)}{1.6 - 1.0} = -0.1087339.$$

The remaining second divided differences are shown in the 5th column of Table 3.11. The third divided difference involving x_0 , x_1 , x_2 , and x_3 and the fourth divided difference involving all the data points are, respectively,

$$\begin{aligned} f[x_0, x_1, x_2, x_3] &= \frac{f[x_1, x_2, x_3] - f[x_0, x_1, x_2]}{x_3 - x_0} = \frac{-0.0494433 - (-0.1087339)}{1.9 - 1.0} \\ &= 0.0658784, \end{aligned}$$

and

$$\begin{aligned} f[x_0, x_1, x_2, x_3, x_4] &= \frac{f[x_1, x_2, x_3, x_4] - f[x_0, x_1, x_2, x_3]}{x_4 - x_0} = \frac{0.0680685 - 0.0658784}{2.2 - 1.0} \\ &= 0.0018251. \end{aligned}$$

All the entries are given in Table 3.11.

The coefficients of the Newton forward divided-difference form of the interpolating polynomial are along the diagonal in the table. This polynomial is

$$\begin{aligned} P_4(x) &= 0.7651977 - 0.4837057(x - 1.0) - 0.1087339(x - 1.0)(x - 1.3) \\ &\quad + 0.0658784(x - 1.0)(x - 1.3)(x - 1.6) \\ &\quad + 0.0018251(x - 1.0)(x - 1.3)(x - 1.6)(x - 1.9). \end{aligned}$$

Divided Differences

Example

Given the data

x_i	$f(x_i)$
1	3
2	6
3	19
5	99

Calculate $f(4)$ using Newton's interpolating polynomials of order 3.

x_i	$f[x_i]$	First	Second	Third
1	3	3		
2	6		5	
3	19	13		1
5	99	40	9	

$$p_3(x) = 3 + 3(x - 1) + 5(x - 1)(x - 2) + 1(x - 1)(x - 2)(x - 3)$$

$$p_3(4) = 48$$

The coefficients of the Newton forward divided-difference form of the interpolating polynomial are along the **diagonal** in the table.

Interpolation with Equally Spaced Data

If data are equally spaced and in ascending order, then the independent variable assumes values of

$$\begin{aligned}x_1 &= x_0 + h \\x_2 &= x_0 + 2h \\&\vdots \\&\vdots \\&\vdots \\x_n &= x_0 + nh\end{aligned}$$

where h is the interval, or step size, between these data. On this basis, the finite divided differences can be expressed in concise form. For example, the second forward divided difference is

$$f[x_0, x_1, x_2] = \frac{\frac{f(x_2) - f(x_1)}{x_2 - x_1} - \frac{f(x_1) - f(x_0)}{x_1 - x_0}}{x_2 - x_0}$$

which can be expressed as

$$f[x_0, x_1, x_2] = \frac{f(x_2) - 2f(x_1) + f(x_0)}{2h^2} \quad (\text{B18.2.1})$$

because $x_1 - x_0 = x_2 - x_1 = (x_2 - x_0)/2 = h$. Now recall that the second forward difference is equal to

$$\Delta^2 f(x_0) = f(x_2) - 2f(x_1) + f(x_0)$$

Therefore, Eq. (B18.2.1) can be represented as

$$f[x_0, x_1, x_2] = \frac{\Delta^2 f(x_0)}{2!h^2}$$

or, in general,

$$f[x_0, x_1, \dots, x_n] = \frac{\Delta^n f(x_0)}{n!h^n} \quad (\text{B18.2.2})$$

Using Eq. (B18.2.2), we can express Newton's interpolating polynomial for the case of equispaced data as

$$\begin{aligned} f_n(x) = & f(x_0) + \frac{\Delta f(x_0)}{h}(x - x_0) \\ & + \frac{\Delta^2 f(x_0)}{2!h^2}(x - x_0)(x - x_0 - h) \\ & + \dots + \frac{\Delta^n f(x_0)}{n!h^n}(x - x_0)(x - x_0 - h) \\ & \dots [x - x_0 - (n - 1)h] + R_n \end{aligned} \quad (\text{B18.2.3})$$

$$R_n = \frac{f^{(n+1)}(\xi)}{(n+1)!}(x - x_0)(x - x_1) \cdots (x - x_n)$$

(Mean Value Theorem)

If $f \in C[a, b]$ and f is differentiable on (a, b) , then a number c in (a, b) exists with

$$f'(c) = \frac{f(b) - f(a)}{b - a}.$$

■

$$f[x_i, x_{i+1}] = \frac{f[x_{i+1}] - f[x_i]}{x_{i+1} - x_i}. \quad (3.8)$$

The Mean Value Theorem applied to Eq. (3.8) when $i = 0$,

$$f[x_0, x_1] = \frac{f(x_1) - f(x_0)}{x_1 - x_0},$$

implies that when f' exists, $f[x_0, x_1] = f'(\xi)$ for some number ξ between x_0 and x_1 . The following theorem generalizes this result.

Theorem Suppose that $f \in C^n[a, b]$ and $x_0, x_1, \dots, x_n$ are distinct numbers in $[a, b]$. Then a number ξ exists in (a, b) with

$$f[x_0, x_1, \dots, x_n] = \frac{f^{(n)}(\xi)}{n!}.$$

■

$$p_n(x) = f[x_0] + f[x_0, x_1](x - x_0) + f[x_0, x_1, x_2](x - x_0)(x - x_1) \\ + \cdots + f[x_0, x_1, \dots, x_n](x - x_0)(x - x_1) \cdots (x - x_{n-1})$$

Newton's divided-difference formula can be expressed in a simplified form when the nodes are arranged consecutively with equal spacing. In this case, we introduce the notation $h = x_{i+1} - x_i$, for each $i = 0, 1, \dots, n - 1$ and let $x = x_0 + sh$. Then the difference $x - x_i$ is $x - x_i = (s - i)h$.

$$P_n(x) = P_n(x_0 + sh) = f[x_0] + sh f[x_0, x_1] + s(s - 1)h^2 f[x_0, x_1, x_2] \\ + \cdots + s(s - 1) \cdots (s - n + 1)h^n f[x_0, x_1, \dots, x_n] \\ = f[x_0] + \sum_{k=1}^n s(s - 1) \cdots (s - k + 1)h^k f[x_0, x_1, \dots, x_k].$$

Using binomial-coefficient notation,

$$\binom{s}{k} = \frac{s(s - 1) \cdots (s - k + 1)}{k!},$$

we can express $P_n(x)$ compactly as

$$P_n(x) = P_n(x_0 + sh) = f[x_0] + \sum_{k=1}^n \binom{s}{k} k! h^k f[x_0, x_1, \dots, x_k].$$

Forward Differences

The **Newton forward-difference formula**, is constructed by making use of the forward difference notation Δ introduced in Aitken's Δ^2 method. With this notation,

$$f[x_0, x_1] = \frac{f(x_1) - f(x_0)}{x_1 - x_0} = \frac{1}{h}(f(x_1) - f(x_0)) = \frac{1}{h}\Delta f(x_0)$$

$$f[x_0, x_1, x_2] = \frac{1}{2h} \left[\frac{\Delta f(x_1) - \Delta f(x_0)}{h} \right] = \frac{1}{2h^2} \Delta^2 f(x_0),$$

and, in general,

$$f[x_0, x_1, \dots, x_k] = \frac{1}{k!h^k} \Delta^k f(x_0).$$

$$P_n(x) = f(x_0) + \sum_{k=1}^n \binom{s}{k} \Delta^k f(x_0)$$

Backward Differences

If the interpolating nodes are reordered from last to first as $x_n, x_{n-1}, \dots, x_0$, we can write the interpolatory formula as

$$\begin{aligned} P_n(x) = & f[x_n] + f[x_n, x_{n-1}](x - x_n) + f[x_n, x_{n-1}, x_{n-2}](x - x_n)(x - x_{n-1}) \\ & + \dots + f[x_n, \dots, x_0](x - x_n)(x - x_{n-1}) \dots (x - x_1). \end{aligned}$$

If, in addition, the nodes are equally spaced with $x = x_n + sh$ and $x = x_i + (s + n - i)h$, then

$$\begin{aligned} P_n(x) &= P_n(x_n + sh) \\ &= f[x_n] + sh f[x_n, x_{n-1}] + s(s+1)h^2 f[x_n, x_{n-1}, x_{n-2}] + \cdots \\ &\quad + s(s+1) \cdots (s+n-1)h^n f[x_n, \dots, x_0]. \end{aligned}$$

Newton Backward-Difference Formula

$$P_n(x) = f[x_n] + \sum_{k=1}^n (-1)^k \binom{-s}{k} \nabla^k f(x_n)$$

$$\nabla p_n = p_n - p_{n-1}, \quad \text{for } n \geq 1, \quad \nabla p_n \text{ (read } n\text{abla } p_n)$$

$$\binom{-s}{k} = \frac{-s(-s-1) \cdots (-s-k+1)}{k!} = (-1)^k \frac{s(s+1) \cdots (s+k-1)}{k!},$$

Centered Differences

The Newton forward- and backward-difference formulas are not appropriate for approximating $f(x)$ when x lies near the center of the table because neither will permit the highest-order difference to have x_0 close to x . A number of divided-difference formulas are available for this case, each of which has situations when it can be used to maximum advantage. These methods are known as **centered-difference formulas**. We will consider only one centered-difference formula, Stirling's method.

For the centered-difference formulas, we choose x_0 near the point being approximated and label the nodes directly below x_0 as $x_1, x_2, \dots$ and those directly above as $x_{-1}, x_{-2}, \dots$. With this convention, **Stirling's formula** is given by

$$\begin{aligned} P_n(x) = P_{2m+1}(x) = & f[x_0] + \frac{sh}{2}(f[x_{-1}, x_0] + f[x_0, x_1]) + s^2 h^2 f[x_{-1}, x_0, x_1] \quad (3.14) \\ & + \frac{s(s^2 - 1)h^3}{2} f[x_{-2}, x_{-1}, x_0, x_1] + f[x_{-1}, x_0, x_1, x_2]) \\ & + \dots + s^2(s^2 - 1)(s^2 - 4) \dots (s^2 - (m - 1)^2)h^{2m} f[x_{-m}, \dots, x_m] \\ & + \frac{s(s^2 - 1) \dots (s^2 - m^2)h^{2m+1}}{2} (f[x_{-m-1}, \dots, x_m] + f[x_{-m}, \dots, x_{m+1}]), \end{aligned}$$

if $n = 2m + 1$ is odd. If $n = 2m$ is even, we use the same formula but delete the last line. The entries used for this formula are underlined in Table 3.13.

x	$f(x)$	First divided differences	Second divided differences	Third divided differences	Fourth divided differences
x_{-2}	$f[x_{-2}]$				
		$f[x_{-2}, x_{-1}]$			
x_{-1}	$f[x_{-1}]$		$f[x_{-2}, x_{-1}, x_0]$		
		$\underline{f[x_{-1}, x_0]}$		$\underline{f[x_{-2}, x_{-1}, x_0, x_1]}$	
x_0	$\underline{f[x_0]}$		$\underline{f[x_{-1}, x_0, x_1]}$		$\underline{f[x_{-2}, x_{-1}, x_0, x_1, x_2]}$
		$\underline{f[x_0, x_1]}$		$\underline{f[x_{-1}, x_0, x_1, x_2]}$	
x_1	$f[x_1]$		$f[x_0, x_1, x_2]$		
		$f[x_1, x_2]$			
x_2	$f[x_2]$				

Example 2 Consider the table of data given in the previous examples. Use Stirling's formula to approximate $f(1.5)$ with $x_0 = 1.6$.

Solution To apply Stirling's formula we use the *underlined* entries in the difference Table 3.14.

Table 3.14

x	$f(x)$	First divided differences	Second divided differences	Third divided differences	Fourth divided differences
1.0	0.7651977				
		−0.4837057			
1.3	0.6200860		−0.1087339		
		<u>−0.5489460</u>		<u>0.0658784</u>	
1.6	<u>0.4554022</u>		<u>−0.0494433</u>		<u>0.0018251</u>
		<u>−0.5786120</u>		<u>0.0680685</u>	
1.9	0.2818186		0.0118183		
		−0.5715210			
2.2	0.1103623				

The formula, with $h = 0.3$, $x_0 = 1.6$, and $s = -\frac{1}{3}$, becomes

$$\begin{aligned} f(1.5) &\approx P_4 \left(1.6 + \left(-\frac{1}{3} \right) (0.3) \right) \\ &= 0.4554022 + \left(-\frac{1}{3} \right) \left(\frac{0.3}{2} \right) ((-0.5489460) + (-0.5786120)) \\ &\quad + \left(-\frac{1}{3} \right)^2 (0.3)^2 (-0.0494433) \\ &\quad + \frac{1}{2} \left(-\frac{1}{3} \right) \left(\left(-\frac{1}{3} \right)^2 - 1 \right) (0.3)^3 (0.0658784 + 0.0680685) \\ &\quad + \left(-\frac{1}{3} \right)^2 \left(\left(-\frac{1}{3} \right)^2 - 1 \right) (0.3)^4 (0.0018251) = 0.5118200. \end{aligned}$$